Logical [Root] New Cluster Move Object Set Tiled Background Viewport

Realtime

NP - 03

Experiments on CISCO PACKET TRACER

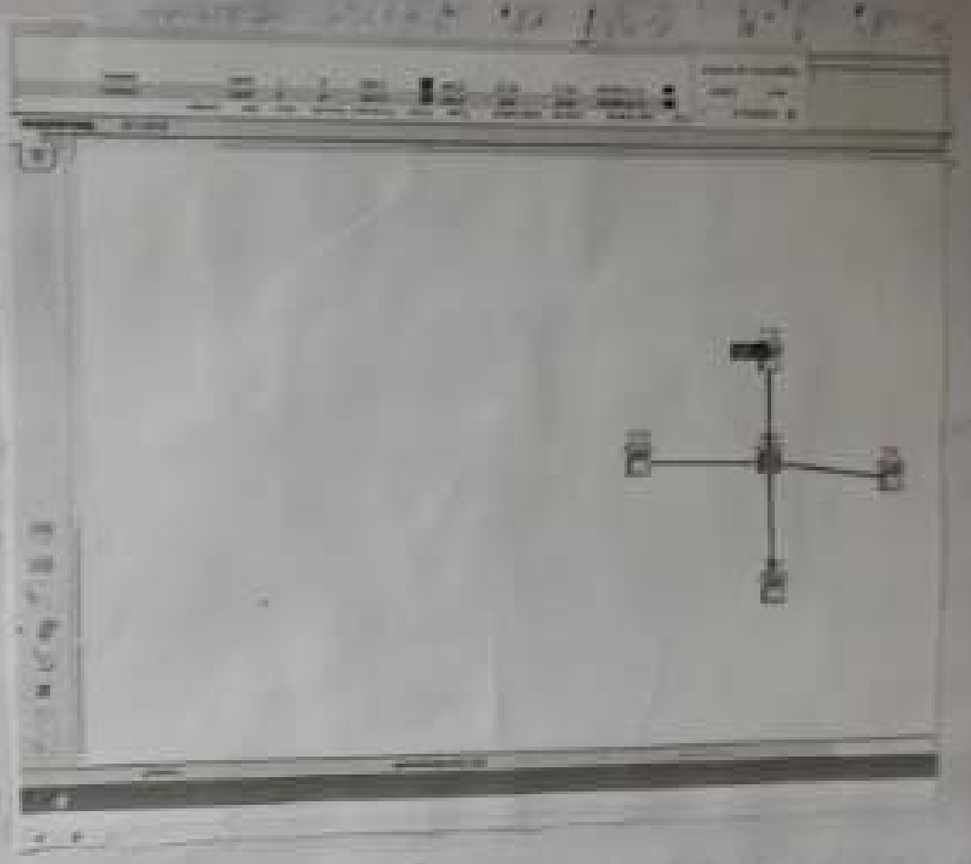
a. Experiment :- understanding the Cisco Packet Tracer Environment.

Aim :-

To understanding the Cisco Packet Tracer environment and design a simple computer network using its interface and networking components.

Steps :-

1. Open the Cisco Packet Tracer application
2. Observe the user interface, including, Common Tools Bar, Main Toolbar, workspace, Common Tools Bar, Network Component box and Realtime / Simulation tabs
3. From the network Component Box, select the Require network devices such as PCs, Hub and switch.
4. Drag and drop the devices into the workspace.
5. Connect the devices using Copper Straight Through Cables.
6. Configure IP addresses after for the PCs using the Desktop → IP Configuration option.
7. Verify that all links are active
8. Observe packet transmission in Realtime mode and Simulator mode.
9. Save the network topology.



Result :-

The CISCO PACKET TRACER environment was successfully explored and simple network was designed.

b Experiment 1 : HUB.

Aim :-

To study the behavior of a Hub by observation how it forwards data packets using cisco packet Tracer.

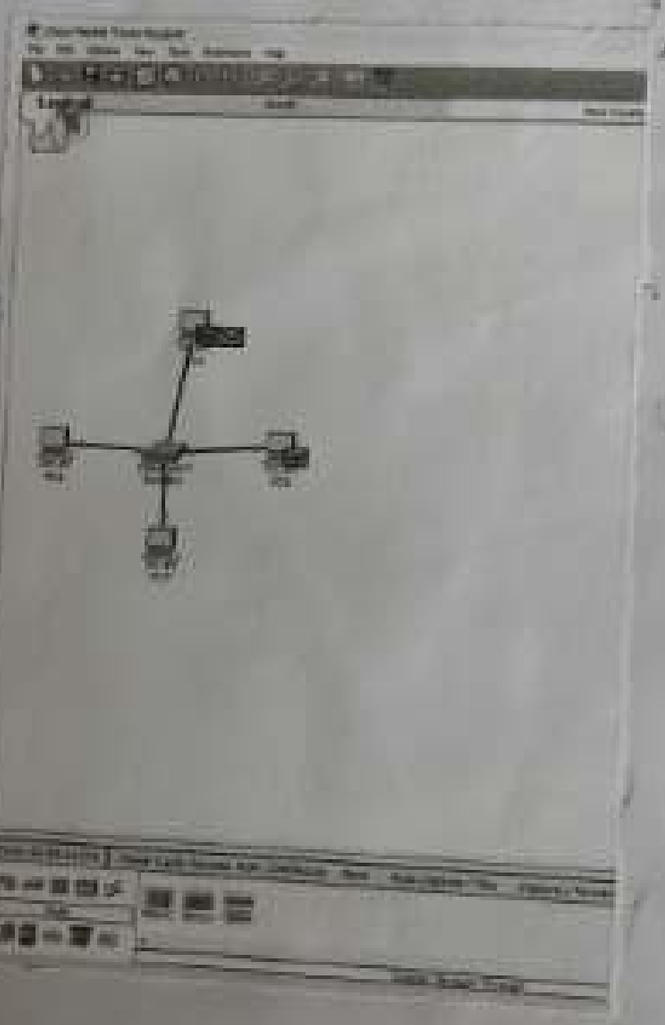
Steps :-

1. Open Cisco packet Tracer.
2. Drag and Place 1 Hub and 4 Generic PC into the workspace
3. Connect all PCs to the Hub using copper straight Through Cables.
4. Configure IP addresses for all PCs in the same network.
5. Select add simple PDU from the common Toolbar
6. Send a packet from one PC to another PC
7. Switch the Simulation Mode and observe the Hub forwards the packet.

Result :-

The Hub broadCasts the received packet to all Connected devices.

SWITCH :



Experiment 2 :- Switch

Aim :-

To Study the behaviour of a switch by observing how it forwards data packets using CISCO PACKET TRACER.

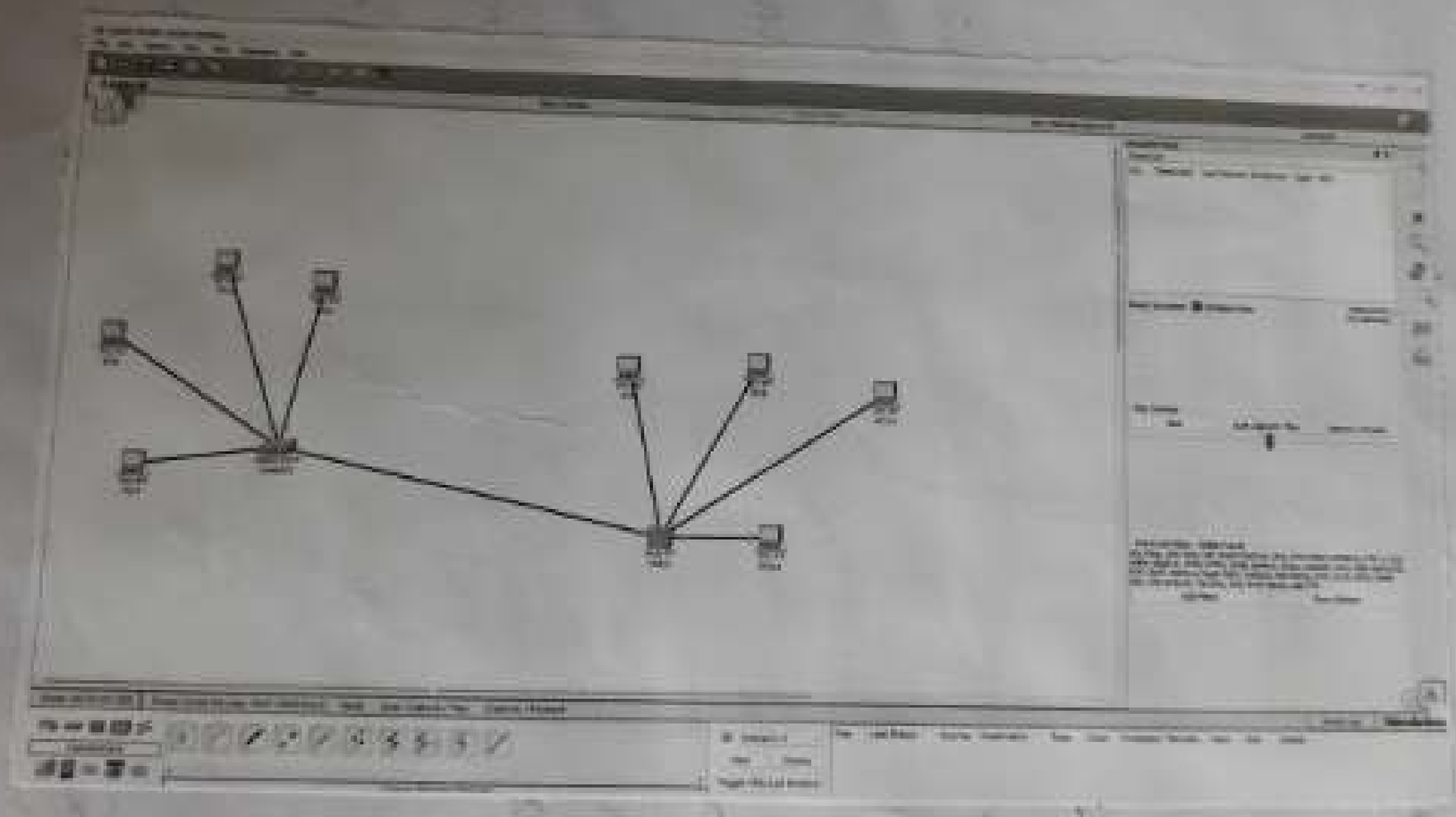
Steps :-

1. Open Cisco Packet Tracer
2. Drag and place 1 Switch and 4 Generic PCs into the workspace.
3. Connect all PCs to the Switch using copper Straight Through Cables
4. Configure IP addresses for all PCs in the same network
5. Select add Simple PDU from the Common Toolbar
6. Send a packet from one PC to another PC
7. Switch to Simulation mode and observe how the Switch forwards the Packet.

Result :-

The Switch forwards the packet only to the destination PC by using its MAC address table.

SWITCH AND HUB



Experiment 3 :- HUB and Switch

Aim :-

To compare the packet forwarding behaviour of a hub and a switch using Cisco packet Tracer.

Steps :-

1. Create one networks using a Hub and another a switch, each connected to 4 PCs.
2. Configure IP addresses for all PCs in both Network
3. Send a simple PDU between two PCs in each Network.
4. Observe packet transmission in Simulation Mode
5. Compare how the hub and switch forward Packets

Results :-

The hub broadcast packets to all connected devices, the switch forwards packets only to the intended destination ; Therefore, reduced network traffic and improved efficiency Compared to the hub.